

MATH 3310 HOMEWORK ASSIGNMENT 1

DUE ON THURSDAY 1 SEPTEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Describe each of the following sets by listing its elements within braces.

$$A = \{n \in \mathbb{N} \mid -4 \leq n < 5\}$$

$$B = \{n \in \mathbb{Z} \mid n^2 < 8\}$$

$$C = \{x \in \mathbb{R} \mid x^2 + 1 = 0\}$$

$$D = \{n \in \mathbb{Z} \mid -4 < n \leq 4\}$$

$$E = \{x \in \mathbb{R} \mid x^2 - 5x + 6 = 0\}$$

$$F = \{n \in \mathbb{Z} \mid 8 < n^3 < 81\}$$

- (2) Consider the sets

$$A = \{n \in \mathbb{Z} \mid 2 \leq |n| < 4\},$$

$$B = \{x \in \mathbb{Q} \mid 2 < x \leq 4\},$$

$$C = \{x \in \mathbb{Q} \mid x^2 - (2 + \sqrt{2})x + 2\sqrt{2} = 0\}, \quad \text{and}$$

$$D = \{x \in \mathbb{R} \mid x^2 - (2 + \sqrt{2})x + 2\sqrt{2} = 0\}.$$

- (a) Describe the set A by listing its elements.
- (b) Determine the set $A \cap B$.
- (c) Give two examples of elements that belong to B and not to A .
- (d) Describe the set D by listing its elements.
- (e) Describe the set C as an intersection of two sets.

- (3) Determine the power set $\mathcal{P}(A)$ and its cardinality $|\mathcal{P}(A)|$ for these sets

$$A = \{a, b\}$$

$$B = \{b, \{c\}, \emptyset\}$$

$$C = \{c, \{d, \emptyset\}, \emptyset\}$$

- (4) Decide whether the following statements are true or false.

- (a) If $\{a\} \in \mathcal{P}(A)$, then $\{a\} \notin A$.
- (b) If X and Y are finite sets with $|Y| = |X| + 1$, then $|\mathcal{P}(Y)| \geq |\mathcal{P}(X)| + 2$.
- (c) If A , B , C , and D are subsets of $\{x, y, z\}$ with $|A| = |B| = |C| = |D| = 2$, then at least two of the four sets are equal.

- (5) Give examples of sets A , B , and C such that $A \neq B$ but $A - C = B - C$.